

HCSM-04

The 14-Mode Multiplet at $\lambda = 4$

The Self-Paired Eigenvalue, the D_4 Orbit Decomposition, and the Support, IPR, and Entropy Classes

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September 2026

Abstract

The Hodge Complex Standard Model identifies the 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus with one Standard Model generation. This paper establishes the spectral content of that multiplet. We prove that the self-paired eigenvalue is $\lambda = 4$ with multiplicity $2(L - 1) = 14$; that the 14 modes decompose under the D_4 point group into three orbits of sizes $(2, 8, 4)$; that the three orbits carry the support classes $\{64, 48, 32\} = \{N, V, N/2\}$; that the inverse participation ratios are $\{2/128, 3/128, 4/128\}$; and that the Shannon entropy classes are $\{6.0, 5.5, 5.0\}$ bits with equal spacing 0.5 bits. The affine entropy-IPR relation $S = \log_2 N + 1 - N \cdot \text{IPR}$ is proved as a theorem for modes whose squared amplitudes take values in $\{0, 1/N, 2/N\}$, and this precondition is verified for the 14 modes of the multiplet. The identification of the 14 modes with one Standard Model generation, and the assignment of an $N/4$ mass label to each mode, are the framework's empirical thesis (HCSM-23) and are stated as disclosed inputs. All numerical claims are verified at machine precision.

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1 Introduction

1.1 The 14-mode multiplet in HCSM

The Hodge Complex Standard Model (HCSM) derives the gauge structure, matter content, particle spectrum, and mixing matrices of the Standard Model from a single discrete substrate: the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. Every downstream result depends on the spectral content of the torus at the self-paired eigenvalue $\lambda = 4$. This paper establishes that spectral content.

1.2 Disclosed inputs

This paper uses the following inputs.

(S1) The substrate is $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with D_4 symmetry (HCSM-01).

(S2) The scalar Laplacian is the 5-point Laplacian

$$(\Delta f)(x_1, x_2) = 4f(x_1, x_2) - f(x_1 + 1, x_2) - f(x_1 - 1, x_2) - f(x_1, x_2 + 1) - f(x_1, x_2 - 1). \quad (1)$$

This is the unique nearest-neighbour, D_4 -invariant, positive-semidefinite operator with kernel equal to the constant functions (HCSM-01, Theorem 2.2.1).

(S3) The substrate size $L = 8$ is selected by equal entropy spacing (HCSM-01, §2.4.1) and by maximal cancellation of the winding generator (HCSM-14.b).

(P1) The 14 modes at $\lambda = 4$ are identified with one Standard Model generation (HCSM-23, §3). This is the framework’s empirical thesis.

(P7) The $N/4$ mass-ordering labels attached to the 14 modes (HCSM-23, Input 3) are empirical anchors.

(P8) The coordinate-to-particle correspondence in Remark 4.2 (HCSM-23).

1.3 Main results

- (1) **Self-paired eigenvalue and multiplicity (Theorem 3.2).** The unique self-paired eigenvalue is $\lambda = 4$ with multiplicity $2(L - 1) = 14$.
- (2) **Explicit mode list (Theorem 4.1).** The 14 modes at $\lambda = 4$.
- (3) **D_4 orbit decomposition (Theorem 5.1).** Three orbits of sizes $(2, 8, 4)$.
- (4) **Orbit stabilizers (Theorem 5.2).** Stabilizer orders $4, 1, 2$.
- (5) **Support classes (Proposition 6.2).** Supports $\{64, 48, 32\}$.
- (6) **Inverse participation ratios (Proposition 7.2).** IPRs $\{2/128, 3/128, 4/128\}$.
- (7) **Affine entropy-IPR relation (Theorem 8.2).** $S = \log_2 N + 1 - N \cdot \text{IPR}$ for modes with squared amplitudes in $\{0, 1/N, 2/N\}$.
- (8) **Entropy classes (Corollary 8.3).** $S \in \{6.0, 5.5, 5.0\}$ bits, equal spacing 0.5 bits.

2 Setup

2.1 The substrate

The substrate is $\Lambda = \mathbb{Z}_L \times \mathbb{Z}_L$ with $N = L^2$ sites. Site (x_1, x_2) is identified with $(x_1 + L, x_2)$ and with $(x_1, x_2 + L)$.

At $d = 4$, $N = d \cdot 2^d = 64$ and $L = \sqrt{N} = 8$.

2.2 The discrete Laplacian

The 5-point Laplacian is

$$(\Delta f)(x) = 4f(x) - \sum_{|e|=1} f(x + e), \quad (2)$$

where the sum runs over $e \in \{(\pm 1, 0), (0, \pm 1)\}$.

2.3 Laplacian eigenvalues

The Laplacian eigenvalue on the complex plane wave $\phi_{n_1, n_2}(x) = e^{2\pi i(n_1 x_1 + n_2 x_2)/L}$ is

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{2\pi n_1}{L}\right) - 2 \cos\left(\frac{2\pi n_2}{L}\right). \quad (3)$$

At $L = 8$, $\lambda(n_1, n_2) = 4 - 2 \cos(\pi n_1/4) - 2 \cos(\pi n_2/4)$. The spectrum is mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$, with 13 distinct eigenvalues.

3 The self-paired eigenvalue

Definition 3.1 (Self-paired eigenvalue). An eigenvalue λ is self-paired if $\lambda = 8 - \lambda$.

Theorem 3.2 (Self-paired eigenvalue and multiplicity). *The unique self-paired eigenvalue of the discrete Laplacian on the 8×8 torus is $\lambda = 4$. Its multiplicity is $2(L - 1) = 2(8 - 1) = 14$.*

Proof. The mirror symmetry $\lambda \leftrightarrow 8 - \lambda$ follows from the substitution $(n_1, n_2) \mapsto (n_1 + 4, n_2)$ or $(n_1, n_2) \mapsto (n_1, n_2 + 4)$, which sends $\cos(\pi n/4) \mapsto -\cos(\pi n/4)$ and hence $\lambda \mapsto 8 - \lambda$. The unique fixed point is $\lambda = 4$.

The eigenvalue $\lambda = 4$ is the set of modes (n_1, n_2) satisfying

$$\cos\left(\frac{\pi n_1}{4}\right) + \cos\left(\frac{\pi n_2}{4}\right) = 0. \quad (4)$$

Using $\cos a + \cos b = 2 \cos \frac{a+b}{2} \cos \frac{a-b}{2}$, this becomes

$$2 \cos\left(\frac{\pi(n_1 + n_2)}{8}\right) \cos\left(\frac{\pi(n_1 - n_2)}{8}\right) = 0. \quad (5)$$

The solutions are $n_1 + n_2 \equiv 4 \pmod{8}$ or $n_1 - n_2 \equiv 4 \pmod{8}$. Each congruence has 8 solutions in $\mathbb{Z}_8 \times \mathbb{Z}_8$; the two congruences intersect in exactly 2 points. By inclusion-exclusion, the total is $2 \cdot 8 - 2 = 14$. $\square$

4 The explicit mode list

Theorem 4.1 (Explicit mode list). *The 14 momenta at $\lambda = 4$ on the 8×8 torus are*

$$\begin{aligned} \mathcal{O}_0 &= \{(0, 4), (4, 0)\}, \\ \mathcal{O}_1 &= \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \\ \mathcal{O}_2 &= \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \end{aligned} \quad (6)$$

The real mode associated to momentum (n_1, n_2) is

$$\varphi_{n_1, n_2}(x_1, x_2) = \frac{\cos\left(\frac{\pi(n_1 x_1 + n_2 x_2)}{4}\right)}{N_{n_1, n_2}}, \quad (7)$$

with N_{n_1, n_2} the normalization $\sum_x |\varphi(x)|^2 = 1$. When (n_1, n_2) is not self-conjugate modulo 8, the companion mode $\psi_{n_1, n_2}(x_1, x_2) = \sin(\pi(n_1 x_1 + n_2 x_2)/4)/N_{n_1, n_2}$ is also an eigenvector at $\lambda = 4$ with the same normalization.

Proof. Direct enumeration of $\cos(\pi n_1/4) + \cos(\pi n_2/4) = 0$ using the values $\cos(\pi n/4) \in \{1, \sqrt{2}/2, 0, -\sqrt{2}/2, -1\}$ for $n \in \{0, \dots, 7\}$. $\square$

Remark 4.2 (Standard mode order and disclosed naming). Throughout HCSM the 14 modes are listed in the standard mode order

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t). \quad (8)$$

The correspondence between the standard mode order and the mode coordinates is a disclosed input (P8):

$$\begin{aligned} \nu &: (0, 4), \quad e : (4, 0), \quad u : (1, 3), \quad d : (1, 5), \\ s &: (3, 1), \quad \mu : (3, 7), \quad \text{dark} : (5, 7), \quad c : (7, 3), \\ \tau &: (7, 5), \quad b : (5, 1), \quad W : (2, 2), \quad Z : (2, 6), \\ H &: (6, 2), \quad t : (6, 6). \end{aligned} \quad (9)$$

5 The D_4 orbit decomposition

5.1 The D_4 action

The D_4 point group acts on Λ by

$$r : (n_1, n_2) \mapsto (-n_2, n_1) \pmod{8}, \quad s : (n_1, n_2) \mapsto (n_1, -n_2) \pmod{8}, \quad (10)$$

subject to $r^4 = s^2 = (rs)^2 = e$. The group has order 8: $D_4 = \{e, r, r^2, r^3, s, rs, r^2s, r^3s\}$.

5.2 The three orbits

Theorem 5.1 (D_4 orbit decomposition). *The 14 modes at $\lambda = 4$ decompose under D_4 into three orbits $\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2$ of sizes 2, 8, 4.*

Proof. Direct computation.

Orbit of $(0, 4)$. $r(0, 4) = (-4, 0) = (4, 0)$; $r^2(0, 4) = (-0, -4) = (0, 4)$. s fixes both. Orbit is $\{(0, 4), (4, 0)\}$, size 2.

Orbit of $(1, 3)$. $r(1, 3) = (-3, 1) = (5, 1)$; $r^2(1, 3) = (-1, -3) = (7, 5)$; $r^3(1, 3) = (3, -1) = (3, 7)$. Reflection coset gives $(1, 5), (7, 3), (5, 7), (3, 1)$. All eight distinct.

Orbit of $(2, 2)$. $r(2, 2) = (-2, 2) = (6, 2)$; $r^2(2, 2) = (-2, -2) = (6, 6)$; $r^3(2, 2) = (2, -2) = (2, 6)$. s swaps within the orbit. Orbit is $\{(2, 2), (2, 6), (6, 2), (6, 6)\}$, size 4. $\square$

Theorem 5.2 (Orbit stabilizers). $|\text{Stab}(\mathcal{O}_0)| = 4$, $|\text{Stab}(\mathcal{O}_1)| = 1$, $|\text{Stab}(\mathcal{O}_2)| = 2$, satisfying $|\mathcal{O}_k| = |D_4|/|\text{Stab}(\mathcal{O}_k)|$.

Proof. Direct computation.

$\mathcal{O}_0$: The stabilizer of $(0, 4)$ is $\{e, r^2, s, r^2s\}$, order 4.

$\mathcal{O}_1$: The stabilizer of $(1, 3)$ is trivial: direct check gives $r(1, 3) = (5, 1)$, $s(1, 3) = (1, 5)$, $r^2(1, 3) = (7, 5)$, $rs(1, 3) = (3, 1)$, $r^2s(1, 3) = (7, 3)$, $r^3s(1, 3) = (5, 7)$, $r^3(1, 3) = (3, 7)$, all distinct from $(1, 3)$. Order 1.

$\mathcal{O}_2$: The stabilizer of $(2, 2)$ is $\{e, rs\}$ since $rs(2, 2) = r(2, -2) = r(2, 6) = (-6, 2) = (2, 2)$, and rs has order 2. Order 2. $\square$

6 Support classes

Definition 6.1 (Support). The support of a normalized mode φ is the number of sites x with $\varphi(x) \neq 0$. The support of a D_4 orbit is the common value of the supports of the orbit's modes.

Proposition 6.2 (Support classes). *The support classes of the three orbits are*

$$\text{supp}(\mathcal{O}_0) = 64 = N, \quad \text{supp}(\mathcal{O}_1) = 48 = V, \quad \text{supp}(\mathcal{O}_2) = 32 = N/2, \quad (11)$$

where $V = (d-1) \cdot 2^d = 3 \cdot 16 = 48$ at $d = 4$.

Proof. We compute the support of each orbit from the explicit mode (7).

$\mathcal{O}_0$. For $(n_1, n_2) = (0, 4)$, the mode is $\varphi(x_1, x_2) = \cos(\pi x_2)/N$. The value $\cos(\pi x_2) = \pm 1$ for every integer x_2 , so φ is nonzero on all 64 sites. The normalization gives $\sum_x \cos^2(\pi x_2) = 64$, so $N = 8$ and $|\varphi|^2 = 1/64$ everywhere. Support = 64. The same holds for $(4, 0)$.

$\mathcal{O}_1$. Both n_1 and n_2 are odd. Define $k(x) = n_1x_1 + n_2x_2 \pmod{8}$. Since n_1, n_2 are coprime to 8, the map $x \mapsto k(x)$ is a surjection onto $\mathbb{Z}_8$, and each residue $k \in \{0, \dots, 7\}$ occurs at exactly 8 sites. Hence

$$N^2 = \sum_x \cos^2\left(\frac{\pi k(x)}{4}\right) = 8 \sum_{k=0}^7 \cos^2\left(\frac{\pi k}{4}\right) = 8 \cdot 4 = 32, \quad (12)$$

where $\sum_{k=0}^7 \cos^2(\pi k/4) = 1 + 1/2 + 0 + 1/2 + 1 + 1/2 + 0 + 1/2 = 4$. So $N = \sqrt{32}$ and

$$|\varphi(x)|^2 = \frac{\cos^2(\pi k(x)/4)}{32}. \quad (13)$$

The value $\cos^2(\pi k/4) = 0$ occurs for $k \in \{2, 6\}$, i.e., at $2 \cdot 8 = 16$ sites. Support = $64 - 16 = 48$.

$\mathcal{O}_2$. Both n_1 and n_2 are even, $n_1 = 2a$, $n_2 = 2b$ with $a, b \in \{1, 3\}$. Then $k(x) = 2(ax_1 + bx_2) \pmod{8}$ is always even. Since a, b are coprime to 4, the map $(x_1, x_2) \mapsto ax_1 + bx_2 \pmod{4}$ is surjective, so each even residue of k occurs at exactly 16 sites. Then

$$N^2 = 16 \sum_{k \in \{0, 2, 4, 6\}} \cos^2\left(\frac{\pi k}{4}\right) = 16 \cdot 2 = 32, \quad (14)$$

where $\sum_{k \text{ even}} \cos^2(\pi k/4) = 1 + 0 + 1 + 0 = 2$. So $N = \sqrt{32}$ and

$$|\varphi(x)|^2 = \frac{\cos^2(\pi k(x)/4)}{32}. \quad (15)$$

The value $\cos^2 = 0$ occurs for $k \in \{2, 6\}$, i.e., at $2 \cdot 16 = 32$ sites. Support = $64 - 32 = 32$. $\square$

7 Inverse participation ratios

Definition 7.1 (Inverse participation ratio). For a normalized mode φ , $\text{IPR}(\varphi) = \sum_x |\varphi(x)|^4$. For a D_4 orbit, the IPR is the common value across the orbit's modes.

Proposition 7.2 (Inverse participation ratios). *The IPRs of the three orbits are*

$$\text{IPR}(\mathcal{O}_0) = \frac{2}{128}, \quad \text{IPR}(\mathcal{O}_1) = \frac{3}{128}, \quad \text{IPR}(\mathcal{O}_2) = \frac{4}{128}. \quad (16)$$

The values are equally spaced by $1/128$.

Proof. We compute each IPR from the amplitude structure derived in Proposition 6.2.

$\mathcal{O}_0$. $|\varphi(x)|^2 = 1/64$ on all 64 sites. Then

$$\text{IPR}(\mathcal{O}_0) = 64 \cdot \left(\frac{1}{64}\right)^2 = \frac{1}{64} = \frac{2}{128}. \quad (17)$$

$\mathcal{O}_1$. $|\varphi(x)|^2 = 1/32$ on 16 sites and $1/64$ on 32 sites. Then

$$\text{IPR}(\mathcal{O}_1) = 16 \cdot \left(\frac{1}{32}\right)^2 + 32 \cdot \left(\frac{1}{64}\right)^2 = \frac{16}{1024} + \frac{32}{4096} = \frac{1}{64} + \frac{1}{128} = \frac{3}{128}. \quad (18)$$

$\mathcal{O}_2$. $|\varphi(x)|^2 = 1/32$ on 32 sites and 0 on 32 sites. Then

$$\text{IPR}(\mathcal{O}_2) = 32 \cdot \left(\frac{1}{32}\right)^2 = \frac{1}{32} = \frac{4}{128}. \quad (19)$$

The equal spacing is $\text{IPR}(\mathcal{O}_0) - \text{IPR}(\mathcal{O}_1) = \text{IPR}(\mathcal{O}_1) - \text{IPR}(\mathcal{O}_2) = 1/128$. $\square$

8 Shannon entropy classes

Definition 8.1 (Shannon entropy). For a normalized mode φ , $S(\varphi) = -\sum_x |\varphi(x)|^2 \log_2 |\varphi(x)|^2$. For a D_4 orbit, the entropy is the common value across the orbit's modes.

Theorem 8.2 (Affine entropy-IPR relation). *Let φ be a normalized mode on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $|\varphi(x)|^2 \in \{0, 1/N, 2/N\}$ for every site x , where $N = 64$. Then the Shannon entropy and the inverse participation ratio satisfy the exact relation*

$$S = \log_2 N + 1 - N \cdot \text{IPR}. \quad (20)$$

Proof. Define $c(x) = N \cdot |\varphi(x)|^2$. The hypothesis is $c(x) \in \{0, 1, 2\}$ for every x .

Normalization gives $\sum_x c(x) = N \cdot \sum_x |\varphi(x)|^2 = N$. The entropy is

$$S = -\sum_x |\varphi(x)|^2 \log_2 |\varphi(x)|^2 = -\frac{1}{N} \sum_x c(x) (\log_2 c(x) - \log_2 N) = \log_2 N - \frac{1}{N} \sum_x c(x) \log_2 c(x). \quad (21)$$

The IPR is

$$\text{IPR} = \sum_x |\varphi(x)|^4 = \frac{1}{N^2} \sum_x c(x)^2. \quad (22)$$

Substituting into the claimed identity gives

$$\log_2 N - \frac{1}{N} \sum_x c \log_2 c = \log_2 N + 1 - \frac{1}{N} \sum_x c^2, \quad (23)$$

which reduces to

$$\sum_x (c(x)^2 - c(x) \log_2 c(x) - c(x)) = 0, \quad (24)$$

using $\sum_x c(x) = N$. Factor the summand as $c(x)(c(x) - \log_2 c(x) - 1)$. For each allowed value:

- $c = 0$: $0 \cdot (\dots) = 0$.
- $c = 1$: $1 \cdot (1 - 0 - 1) = 0$.
- $c = 2$: $2 \cdot (2 - 1 - 1) = 0$.

The sum vanishes term-by-term. Hence the affine relation holds. $\square$

Corollary 8.3 (Entropy classes). *The Shannon entropy classes of the three orbits are*

$$S(\mathcal{O}_0) = 6.0 \text{ bits}, \quad S(\mathcal{O}_1) = 5.5 \text{ bits}, \quad S(\mathcal{O}_2) = 5.0 \text{ bits}, \quad (25)$$

with equal spacing 0.5 bits.

Proof. We verify that every mode in the multiplet satisfies the hypothesis of Theorem 8.2, then apply the relation.

$\mathcal{O}_0$. From Proposition 6.2, $|\varphi|^2 = 1/64 = 1/N$ everywhere. Hypothesis holds. Then $S = \log_2 64 + 1 - 64 \cdot (2/128) = 6 + 1 - 1 = 6.0$.

$\mathcal{O}_1$. $|\varphi|^2 \in \{0, 1/32, 1/64\} = \{0, 2/N, 1/N\}$. Hypothesis holds. Then $S = \log_2 64 + 1 - 64 \cdot (3/128) = 6 + 1 - 1.5 = 5.5$.

$\mathcal{O}_2$. $|\varphi|^2 \in \{0, 1/32\} = \{0, 2/N\}$. Hypothesis holds. Then $S = \log_2 64 + 1 - 64 \cdot (4/128) = 6 + 1 - 2 = 5.0$.

The equal spacing follows: $\Delta S = N \cdot \Delta \text{IPR} = 64 \cdot (1/128) = 0.5$ bits. $\square$

Remark 8.4 (Consistency with the equal-entropy-spacing anchor). The entropy classes $\{6.0, 5.5, 5.0\}$ bits are the three equally-spaced classes selected by the equal-entropy-spacing anchor of HCSM-01. Among all integers $4 \leq L \leq 10^4$, only $L = 8$ produces exactly three equally-spaced entropy classes.

9 The Hodge lift

The 14 modes at $\lambda = 4$ lift to the full Hodge complex as

$$14\Omega^0 + 28\Omega^1 + 14\Omega^2 = 56. \quad (26)$$

The D_4 content of the 56-dimensional sector is $9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E$, derived in HCSM-02.

10 Disclosed inputs and open items

10.1 Derived

All results in Sections 3.2 through the entropy corollary are proved in this paper from (S1)-(S3):

1. The self-paired eigenvalue $\lambda = 4$ and its multiplicity 14 (Theorem 3.2).
2. The explicit mode list (Theorem 4.1).
3. The D_4 orbit decomposition $(2, 8, 4)$ (Theorem 5.1).
4. The stabilizer orders $4, 1, 2$ (Theorem 5.2).
5. The support classes $\{64, 48, 32\}$ (Proposition 6.2).
6. The IPRs $\{2/128, 3/128, 4/128\}$ (Proposition 7.2).
7. The affine entropy-IPR relation (Theorem 8.2).
8. The entropy classes $\{6.0, 5.5, 5.0\}$ bits (Corollary 8.3).

10.2 Disclosed inputs

(S1)-(S3) Substrate, Laplacian, and $L = 8$ selection (HCSM-01, HCSM-14.b).

(P1) Identification of the 14 modes with one Standard Model generation (HCSM-23, §3).

(P7) $N/4$ mass-ordering labels (HCSM-23, Input 3).

(P8) Coordinate-to-particle correspondence (HCSM-23).

10.3 Open items

The identification thesis (P1) and the $N/4$ labels (P7) are not derived in this paper or in any HCSM paper. They are the framework's empirical anchors.

11 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script `HCSM-04_verification.py` constructs the 14 modes from (7), computes their amplitudes, supports, IPRs, and Shannon entropies, and verifies:

1. The self-paired eigenvalue $\lambda = 4$ and its multiplicity 14 (Theorem 3.2).
2. The explicit mode list of Theorem 4.1.

3. The D_4 orbit decomposition (Theorem 5.1).
4. The orbit stabilizer orders (Theorem 5.2).
5. The support classes $\{64, 48, 32\}$ (Proposition 6.2).
6. The IPRs $\{2/128, 3/128, 4/128\}$ (Proposition 7.2).
7. The amplitude-value precondition $|\varphi|^2 \in \{0, 1/N, 2/N\}$ (Theorem 8.2).
8. The affine entropy-IPR relation (Theorem 8.2).
9. The entropy classes $\{6.0, 5.5, 5.0\}$ bits (Corollary 8.3).
10. The equal spacing 0.5 bits.
11. The Hodge lift $14 + 28 + 14 = 56$.

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

12 Conclusion

This paper has established the 14-mode multiplet at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus. The self-paired eigenvalue has multiplicity $2(L - 1) = 14$. The 14 modes decompose under D_4 into three orbits of sizes $(2, 8, 4)$ with stabilizer orders 4, 1, 2.

The three orbits carry the support classes $\{64, 48, 32\} = \{N, V, N/2\}$, the inverse participation ratios $\{2/128, 3/128, 4/128\}$, and the Shannon entropy classes $\{6.0, 5.5, 5.0\}$ bits with equal spacing 0.5 bits. The affine entropy-IPR relation $S = \log_2 N + 1 - N \cdot \text{IPR}$ is proved as a theorem for modes whose squared amplitudes lie in $\{0, 1/N, 2/N\}$, and this precondition is verified for every mode in the multiplet.

The identification of the 14 modes with one Standard Model generation, and the assignment of an $N/4$ mass label to each mode, are the framework's empirical thesis (HCSM-23).

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